

1. Administrative & Study Identification

Official Study Title: A Phase 3, Randomized, Double-Blind Study of Nivolumab or Placebo in Combination With Docetaxel, in Men With Metastatic Castration-resistant Prostate Cancer **Registry Study Title:** A Study of Nivolumab or Placebo in Combination With Docetaxel in Men With Advanced Castration-resistant Prostate Cancer **Short Title/Acronym:** CheckMate 7DX **Protocol Number:** CA209-7DX **NCT Number:** NCT04100018 **Posting ID:** 860721238244589839 **Sponsor/Company Name:** Bristol Myers Squibb **Investigational Product:** Nivolumab in combination with Docetaxel and Prednisone **Phase of Development:** PHASE3 **Medical Monitor:** Clinical Operations Lead, Bristol Myers Squibb

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate Radiographic Progression-Free Survival (rPFS) and Overall Survival (OS) of nivolumab in combination with docetaxel versus placebo in combination with docetaxel in men with advanced castration-resistant prostate cancer who have progressed after second-generation hormonal manipulation. **Secondary Objectives:** To evaluate Objective Response Rate (ORR), Duration of Response (DOR), Time to Response (TTR), PSA Response Rate (PSA-RR), and safety.

2.2 Background & Rationale Androgen receptor pathway inhibitors (ARPIs) and docetaxel are established standards of care for chemotherapy-naïve metastatic castration-resistant prostate cancer (mCRPC). This study aims to evaluate the clinical benefit, long-term safety, and survival impact of adding nivolumab, an anti-PD-1 immune checkpoint inhibitor, to standard-of-care docetaxel chemotherapy versus docetaxel alone in ARPI-pretreated, chemotherapy-naïve mCRPC patients.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Trial Status:** COMPLETED **Control Type:** Placebo-controlled, Active Comparator (Docetaxel + Prednisone) **Blinding:** Double-blind **Randomization:** 1:1, Stratified by previous ARPI therapy and visceral disease

3.2 Planned Enrollment & Duration Targeted Enrollment: 1030 randomly assigned **Number of Sites:** 291 hospitals and cancer centres across 27 countries **Estimated Study Start:** 11-Mar-2020 **Estimated Primary Completion:** 01-Jun-2023

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adult men (aged ≥ 18 years) with histologic confirmation of adenocarcinoma of the prostate without small cell features. 2. Current evidence of metastatic disease documented by either bone lesions on radionuclide bone scan and/or soft tissue lesions on computerized tomography/magnetic resonance imaging (CT/MRI). 3. Eastern Cooperative Oncology Group (ECOG) performance status 0-1. 4. Ongoing androgen deprivation therapy (ADT) with a gonadotropin-releasing hormone (GnRH) analogue or bilateral orchiectomy. 5. Documented prostate cancer progression per Prostate Cancer Working Group (PCWG3) criteria within 6 months prior to screening. 6. Chemotherapy-naïve for metastatic castration-resistant prostate cancer (mCRPC), with 1 to 2 prior second generation hormonal therapies in the recurrent non-metastatic setting and/or metastatic setting, and no more than 1 second generation hormonal therapy in the mCRPC setting. Must have progressed during or after second generation hormonal therapy or have documented intolerance to second generation hormonal therapy. 7. Participants must meet one of the following criteria regarding tissue submission: Sufficient tumor samples from a newly obtained ("fresh") biopsy (obtained during screening); or archival tumor tissue in the form of formalin-fixed paraffin-embedded (FFPE) block or unstained tumor tissue slides. For participants with bone-only disease or inaccessible soft tissue lesions or if the biopsy procedure would pose an unacceptable clinical risk for the participant, submission of tumor tissue obtained from a fresh biopsy is not required. 8. Men must agree to follow specific methods of contraception, if applicable.

4.2 Exclusion Criteria 1. Active brain metastases. 2. Active, known, or suspected autoimmune disease. 3. Condition requiring systemic treatment with corticosteroids (> 10 mg daily prednisone equivalent) or other immunosuppressive medications within 14 days of start of study treatment. Inhaled or topical steroids or adrenal replacement steroid doses are permitted in the absence of active autoimmune disease. 4. Prior treatment with docetaxel or other chemotherapy for mCRPC. Prior docetaxel for metastatic castration-sensitive prostate cancer is permitted if at least 12 months have elapsed from last dose of docetaxel. 5. Prior treatment with an anti-PD-1, anti-PD-L1, anti-PD-L2, anti-CTLA-4 antibody, or any other antibody or drug specifically targeting T-cell co-stimulation or checkpoint pathways.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Nivolumab (360 mg) or equivalent placebo and docetaxel (75 mg/m²) intravenously every 3 weeks for up to 10 doses, followed by nivolumab (480 mg) or placebo every 4 weeks. **Additional Drug Information:** Prednisone

(ATC Codes: A07EA03, H02AB07; Associated Trade Names: Rayos, Predone, Prednisone Intensol, Econosone, Prednicot, Liquid Pred, Meticorten, Orasone, Decortisyl, Deltasone, Prednicen-M, Sterapred DS, Sterapred). **Route of Administration:** Intravenous (IV) and Oral. **Duration of Treatment:** Until disease progression, unacceptable toxicity, or completion of protocol-specified maximum treatment duration.

5.2 Concomitant & Prohibited Therapy Permitted: Inhaled or topical steroids, adrenal replacement steroid doses in the absence of active autoimmune disease, and ongoing ADT (GnRH analogue or bilateral orchiectomy). **Prohibited:** Systemic corticosteroids (> 10 mg daily prednisone equivalent) or other immunosuppressive medications within 14 days of study treatment start; other experimental anti-cancer therapies.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Imaging (CT/MRI/Bone scans), Tissue biopsy
Baseline	Day 0	Randomization, First Dose
Interim	Every 3 to 4 Weeks	Treatment administration, Safety Labs, Efficacy Assessments (Tumor scans per PCWG3)
End of Study	Variable	Safety Follow-up (100 days post-last dose), Survival Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints

- 1. Radiographic Progressive Free Survival (rPFS):** Assessed by Blinded Independent Central Review (BICR) Per Prostate Cancer Working Group 3 (PCWG3). Defined as the time between randomization and the first date of documented progression or death due to any cause. Radiographic progression includes at least a 20% relative increase and 5 mm absolute increase in the sum of diameters of target lesions, or the appearance of new lesions. Censoring logic applied for missed assessments or alternate therapies.
- 2. Overall Survival (OS):** Time between randomization and the date of death from any cause. Censored at the last date the participant was known to be alive or at randomization if no follow-up.

7.2 Secondary & Exploratory Endpoints

- 1. Objective Response Rate (ORR):** Assessed by Blinded Independent Central Review (BICR) Per PCWG3.
- 2. Time to Response (TTR):** Assessed by Blinded Independent Central Review (BICR) Per PCWG3.
- 3. Duration of Response:** Assessed by Blinded Independent Central Review (BICR) Per PCWG3.
- 4. PSA Response Rate:** PSA-RR, defined as $\geq 50\%$ decrease from baseline.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Graded per NCI CTCAE v5.0. Specific focus on immune-mediated adverse events (IMAEs) monitored throughout the study and followed up post-treatment. **Serious Adverse Events (SAEs):** Continuously reported; safety assessed in all patients receiving at least one dose of the study drug. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee convened for regular review of safety and efficacy data.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy analyses (all randomly assigned patients). Safety analysis set included all patients who received at least one dose of study treatment. **Handling of Missing Data:** Censoring rules applied for rPFS at the last radiographic tumor assessment up to the start of subsequent cancer therapy for those without progression or death.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study was conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Rigorous onsite and remote monitoring with data verification. **Audit Trail:** Robust Electronic Data Capture (EDC) locking, central imaging verification by BICR.

11. Study Identifiers & Governance

Posting ID: 860721238244589839 **Primary ID:** CA209-7DX **Registry Number:** NCT04100018 **Sponsor/Lead Organization:** Bristol Myers Squibb **Study Title:** CheckMate 7DX **Registry Study Title:** A Study of Nivolumab or Placebo in Combination With Docetaxel in Men With Advanced Castration-resistant Prostate Cancer **Official Regulatory Title:** A Phase 3, Randomized, Double-Blind Study of Nivolumab or Placebo in Combination With Docetaxel, in Men With Metastatic Castration-resistant Prostate Cancer

12. Clinical Framework (Oncology Specific)

Disease Areas / Conditions: Prostate Cancer (Metastatic Castration-Resistant Prostate Cancer) **Diagnosis Threshold:** Histologic confirmation of adenocarcinoma without small cell features, disease progression per PCWG3 after second-generation hormonal therapy **Medical Methodology:** Docetaxel Chemotherapy + Anti-PD-1 Immunotherapy **Optimization Target:** Delay in radiographic progression and improvement in overall survival

13. Study Procedures & Methodology

Management Pathway: Standard clinical pathway for mCRPC combined with experimental immunotherapy dosing. **Education Protocol (HCP):** RECIST and PCWG3 criteria training, immune-mediated AE management guidelines. **Education Protocol (Patient):** Immunotherapy side-effects education and reporting guidelines. **Quality Audit Mechanism:** Blinded Independent Central Review (BICR) for all efficacy scans to ensure objectivity.

14. Observation & Follow-Up Schedule

Total Duration: Follow-up until death, study withdrawal, or study termination. **Onsite Visit Cadence:** Every 3 weeks during combination therapy, followed by every 4 weeks during monotherapy maintenance. **Remote Monitoring Frequency:** Continuously supported via standard pharmacovigilance follow-ups. **Checklist Review Interval:** Regular radiographic assessments aligned with PCWG3 standards.

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				